The Tool Switching Problem

Rafael Colares[a], Mauricio de Souza[b], Annegret Wagler[a]

[a] *Université Clermont Auvergne, Clermont Auvergne INP, CNRS, LIMOS, Clermont-Ferrand, 63000, France*
[b] *Universidade Federal de Minas Gerais, Belo Horizonte, MG, Brazil*

Abstract

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1. Problem Description

The Tool Switching (TS) problem can be described as follows. We are given a set of jobs $J = \{j_1, j_2 \ldots, j_n\}$ to be carried within a fixed timeframe (say, a working shift). A set of tools $T = \{t_1, t_2, \ldots, t_m\}$ is available; and each job $j_i \in J$ requires a subset of tools $T_i \subseteq T$ in order to be fulfilled. In order to carry the jobs, a tool repository of capacity B is at disposal. The tool repository can therefore host any subset of tools $C \subseteq T$ as long as $|C| \leq B$. The subset of tools C inside the tool repository is hereafter called a *configuration*. Without loss of generalization, we can consider that the tool repository is always full, *i.e.*, $|C| = B$. A job $j_i \in J$ can only be carried if all the required tools appear in the tool repository, that is, if $T_i \subseteq C$. We also assume that the tool repository has enough capacity to carry all jobs, since otherwise the problem is clearly unfeasible. In other words, for any job $j_i \in J$, we have $|T_i| \leq B$.

If C does not contain all the required tools to carry a job $j_i \in J$, then tool switching operations are needed. A tool switching operation consists of replacing an unnecessary tool $t_j \in C \setminus T_i$ in the repository by a required tool $t_i \in T_i \setminus C$. In order to carry j_i, this operation needs to be done for all tools in $T_i \setminus C$. The goal is to carry all jobs in a way that the number of tool switching operations is minimized. This corresponds to finding an ordering of the jobs that requires the minimum total the number of tool switching operations.

2. A combinatorial interpretation of TS

Let $\mathcal{C} = \{C_1, C_2, \ldots, C_p\}$ be the collection of all possible tool repository configurations. Let us consider the complete graph K_p, where each vertex corresponds to a configuration. Let $d : E(K_p) \to \mathbb{R}$ be a distance function between two given configurations. In this sense, $d(i, j)$ denotes the number of tool switches required to obtain configuration C_j from configuration C_i, that is,

$$d(i, j) = b - |C_i \cap C_j|.$$

Let $H_j \subseteq V(K_p)$, denote the subset (cluster) of configurations that are capable of carrying job $j \in J$. That is,

$$H_j = \{C_i \in \mathcal{C} : T_j \subseteq C_i\}.$$

Solving the Tool Switching problem can now be seen as the problem of finding the shortest Hamiltonian cycle in K_p visiting a collection of vertices $X \subseteq V(K_p)$ with the property that at least one vertex from each cluster H_j, for $j \in J$, is visited (*i.e.*, is contained in X). We can thus identify two NP-Hard problems in the Tool Switching Problem:

- Covering Problem: Clusters H_j must be covered by X.
- Traveling Salesman Problem: Once the covering is defined, we need to compute a TSP to define the best ordering of jobs.

The tool switching problem can thus be interpreted as combination of Covering and TSP problems.

3. The Tool Switching Problem and Generalized Traveling Salesman Problem

The Generalized Traveling Salesman Problem (GTSP) is a well-known combinatorial problem, see Pop et al. (2024). It can be defined as follows. Given an undirected graph G whose vertices are divided into a given number of clusters, denoted by $V_1, ..., V_n$, the GTSP searches for the shortest Hamiltonian cycle visiting a collection of vertices with the property that exactly one vertex from each cluster is visited.

In this sense, GTSP can be seen as a particular case of Tool Switching Problem where the clusters H_j are all disjoint. [1]

[1] Can we provide a property of the jobs that would induce disjoint clusters? I think it is the case when $|T_i \cup T_j| > B$ for every $j_i, j_j \in J$

4. A non-compact formulation for the Tool Switching Problem

Let $y(v)$ denote a binary variable stating whether or not configuration $v \in V(K_p)$ is used to carry some job. Let $x(u,v)$ denote a binary variable stating whether or not configuration $v \in V(K_p)$ is applied right after configuration $u \in V(K_p)$. The Tool Switching Problem can thus be modeled as follows.

$$\min \sum_{i \in V} \sum_{j \in V} d(u,v) x(u,v) \tag{1}$$

subject to

$$\sum_{v \in H_j} y(v) \geq 1 \qquad \forall j \in J, \tag{2}$$

$$\sum_{u \in V} x(u,v) = y(v) \qquad \forall v \in V, \tag{3}$$

$$\sum_{u \in V} x(v,u) = y(v) \qquad \forall v \in V, \tag{4}$$

$$\sum_{u \in H(S)} \sum_{v \in H(S)} x(u,v) \leq \sum_{v \in H(S)} y(v) - 1 \qquad \forall S \subseteq J, 2 \leq |S| \leq |J| - 1, \tag{5}$$

$$x(u,v) \in \{0,1\} \qquad \forall u \in V, v \in V, \tag{6}$$

$$y(v) \in \{0,1\} \qquad \forall v \in V, \tag{7}$$

where $H(S) = \bigcup_{j \in S} H_j$.

Lemma 1. *The distance function d satisfies the triangular inequalities. That is,*

$$d(u,v) \leq d(u,w) + d(w,v)$$

for any three vertices u, w, v in V.

The formulation described above is clearly non-compact as there are exponentially many possible configurations. Indeed, the number of possible configurations is given by

$$p = \binom{m}{b} = \frac{m!}{b!(m-b)!}.$$

Therefore the size of V is exponential in the size of T.

5. A column-generation approach for the Tool Switching Problem

The formulation described by (1)-(7) is therefore intractable even for small instances. In order to tackle larger instances, we propose a column-generation approach. For this, let us consider the following Restricted Master Problem (RMP), where only a subset of configurations $\bar{V} \subseteq V$ is available.

$$\min z = \sum_{i \in \bar{V}} \sum_{j \in \bar{V}} d(u,v)x(u,v) \tag{8}$$

subject to

$$\sum_{v \in \bar{V} \cap H_j} y(v) \geq 1 \qquad \forall j \in J, \tag{9}$$

$$\sum_{u \in \bar{V}} x(u,v) = y(v) \qquad \forall v \in \bar{V}, \tag{10}$$

$$\sum_{u \in \bar{V}} x(v,u) = y(v) \qquad \forall v \in \bar{V}, \tag{11}$$

$$\sum_{u \in \bar{V} \cap H(S)} \sum_{v \in \bar{V} \cap H(S)} x(u,v) \leq |S| - 1 \qquad \forall S \subseteq J, 2 \leq |S| \leq |J| - 1, \tag{12}$$

$$x(u,v) \in \{0,1\} \qquad \forall u \in \bar{V}, v \in \bar{V}, \tag{13}$$

$$y(v) \in \{0,1\} \qquad \forall v \in \bar{V}, \tag{14}$$

Notice that we can easily ensure the feasibility of the RMP by considering a preprocessing step where a small set of configurations is generated.[2] Given a solution (x, y) of (8)-(12), the column generation step consists of finding a new configuration in $V \setminus \bar{V}$ that would allow the improvement of the objective function z. This can be done by solving a pricing problem which can be stated as follows.

[2] Preprocessing step to be written.

$$\max w = ???? \tag{15}$$

subject to

$$\sum_{t \in T} x(t) = b \tag{16}$$

$$y(i) \leq x(t) \qquad \forall j_i \in J, t \in T_i \tag{17}$$

$$x(t) \in \{0, 1\} \qquad \forall t \in T, \tag{18}$$

$$y(i) \in \{0, 1\} \qquad \forall j_i \in J, \tag{19}$$

The objective function of the pricing subproblem uses the dual variables associated to the constraints in the RMP. To be written.

References

Pop, P.C., Cosma, O., Sabo, C., Sitar, C.P., 2024. A comprehensive survey on the generalized traveling salesman problem. European Journal of Operational Research 314, 819–835.